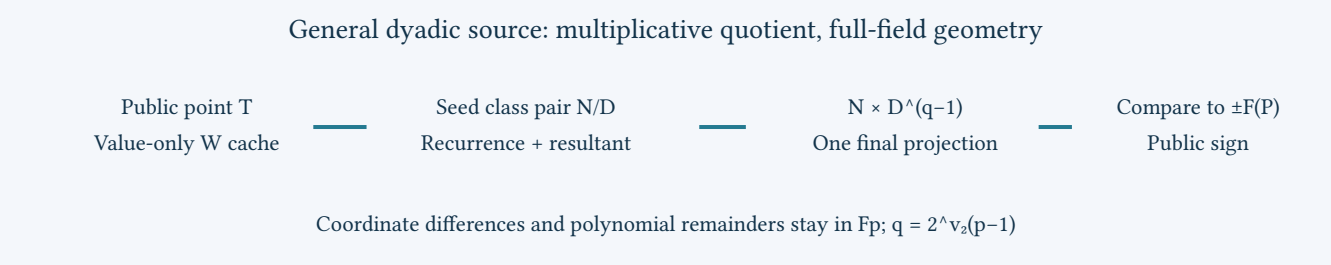
General dyadic quotient-class source

T-1824 · 2026-11-07 · derived algebra and native producer checks

The public sign source now carries multiplicative classes for arbitrary dyadic q, including numerator/denominator pairs throughout its seed and block recurrence. It retains the prior projected source as a full-value reference. Prime p and odd prime subgroup order r are caller-certified; gcd(r,p−1)=1, r≠p, word-size and geometry-lane bounds remain explicit.



Derived invariant. The quotient the multiplicative group modulo q-th powers has exponent q. Nonzero powers reduce modulo q, and a class inverse is its (q−1)-st power. Zero powers are handled before reducing exponents. The resultant accumulator uses this quotient while all Euclidean remainders and their actual degrees remain in Fp. For beta=$W(r+1)^2W(2)/W(r+2)$, $e=(r^2)^{-1}$ mod q, the seed pair is $(W(r+2)^e, (W(r+1)^2W(2))^e)$. Multiplicative recurrence steps commute with projection. Calibration compares raw output to the base value or its negative.

Exact check	Count	Evidence
Class multiplication pairs	114,751	validation-02/check.stdout
Projected root resultants	1,792	Seven prime fields, degrees 1–16
Source points	1,108	Full reference, direct products, covariance, signs
Identity-crossing windows	24	Prepared window checks
New-domain recovery targets	617	5,929 source calls; zero unresolved
Inherited public comparisons	63	511 source calls

Controls. Projection orders include 8,16,32,64,256. Large exponents, zero inversion and cross-field combination are checked. The wrong exponent e=1 changes 266 point-function values. The initial passing source and generator are preserved under validation-01; validation-02 follows removal of obsolete copied test includes. Final native source and recovery builds pass; warnings remain in the receipts. A normal metadata build has five dependencies and excludes derivative jets and prior source implementations.

Costs. Class powers/inverse cost O(log q) field multiplications; final projection costs O(log((p−1)/q)). Seed formation, seed inversion, final source division and calibration no longer require their full-field inverses. Geometry, division-polynomial cache and remainder arithmetic still have inversion costs. Classes store field representatives, not short logarithm encodings. Square-root geometry remains. No timing comparison or cloud launch; original 4,563-case cohort unchanged.

Evidence and next step. PCC-20261107-8013b5 is an unsigned producer statement, not an independent signed verdict. See DERIVATION.md and receipts for assumptions, commands, hashes and replay. Prior Ward normalization uses Lauter–Stange2008; FLINT half-GCD documentation informed field-valued remainder handling (generated-prose notice recorded). KB unavailable. Next account total field operations against the full representative route before selecting a runtime backend.